



Action Learning Literature Review Outline

Layer-wise Attention Entropy for Real-Time Hallucination Detection in Small LLMs: A Comparative Study Across General, Medical, and Legal Domains

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1. Problem Context:

Problem being Addressed:

A phenomena called as hallucination occurs when large language models (LLMs) provide outputs that are confident, fluent, and occasionally factually incorrect. The issue is architectural: instead of referencing verified facts LLMs use learned statistical patterns to create the next token. The model nonetheless generates a response when a pattern is unclear or nonexistent, even if it might not be supported by reality (Rawte et al., 2024). Small models with 1 to 3 billion parameters represent a particular and unexplored dimension. Although these models are frequently utilized in settings with limited resources, research on hallucination detection has mostly concentrated on 7B+ parameter models (Chuang et al., 2024a; Chen et al., 2024; Binkowski et al., 2025). It is completely unknown if detection signals generalize to small models.

Cross-domain transmission is another unsolved issue. When it comes to medical or legal terminology a detector that was trained on general QA data might not function consistently. Even GPT-4o gets an F1 as low as 0.625 on the most challenging medical hallucination instances, as Pandit et al. (2025) showed. According to Hou et al. (2024), hallucinations are present in about 80% of legal assessments produced by LLM. The transfer of attention-based detectors trained on general QA to these specialized domains at 1B–3B scale has not been investigated.

Importance:

In clinical decision support, a hallucinated drug interaction or diagnostic recommendation can harm patients. In legal practice, a fabricated case citation can mislead practitioners. These are the exact settings where small, efficient LLMs are deployed and where the existing detection research offers the least coverage. This project fills that gap for three model architectures, three evaluation domains and three families of detection methods.

2. Overview of Existing Work & Comparison of Approaches

Sahoo et al. (2024) divide the hallucination detection literature into three families of methods. This review adheres to that taxonomy, as it maps directly onto the three experimental tasks that this project implements.

Family	Key Papers	Strengths	Weaknesses
Consistency-based	Su et al. (2024)	No model internals, no annotation, real-time via MIND	Multiple forward passes (K=5–10), high latency, not validated on sub-3B models
Output-entropy	Farquhar et al. (2024); Nikitin et al. (2024)	Statistically grounded, proven, strong results on large models	Requires K sampled responses, fails on confident hallucinations, slow
Attention / internal-state	Chuang et al. (2024a); Chen et al. (2024); Binkowski et al. (2025); Han et al. (2025)	Single forward pass, interpretable, current SOTA for attention-based detection	All prior work on 7B+ models only, cross-domain transfer not studied

The main point of this project is that only internal-state methods can really detect in real-time. Janiak et al. (2025) have raised a concern that detection results are systematically over-reported when using ROUGE evaluation - methods can drop up to 45.9% when evaluated properly. This is supported by Kulkarni et al. (2025) a rigorous evaluation should include multiple, complementary metrics. Chuang et al. (2024b) show in DoLa that factual knowledge is not evenly distributed across the layers of transformer models but is instead concentrated in certain mid-to-upper layers. This justifies the layer-wise analysis theoretically.

3. Identified Gaps and Project Positioning:

Gap 1 - The small-model regime is unstudied. All attention-based and internal state detection papers (Lookback Lens, INSIDE, Spectral Attention Maps, Simple Factuality Probes) tested only on 7B+ models. We have no idea if these signals exist, persist or degrade at 1B-3B scale.

Gap 2 - No layer-wise entropy analysis. Chuang et al. (2024a) and Binkowski et al. (2025) did not ask which particular transformer layer contains the most discriminative entropy signal. DoLa shows that factual knowledge is layer-localised but no one has translated this to a layer-specific detection protocol.

Gap 3 - Small scale cross-domain transfer has not been explored. No paper has tested zero-shot transfer of a general QA trained probe to medical and legal domains on 1B-3B models, nor quantified the performance degradation.

This project bridges all three gaps simultaneously by exploring layer-wise Shannon attention entropy as a hallucination detection signal - profiling entropy at every transformer layer on Llama-3.2-1B, Llama-3.2-3B, and Qwen2.5-1.5B - the first study at this scale. We consider three independent tasks that focus on a single family of methods, using the same models, datasets and evaluation protocols (AUROC, F1, ECE, latency) as recommended in Janiak et al. (2025) and Kulkarni et al. (2025).

4. Research To Application Mapping:

Research Insight	Source	How We Apply It
Attention ratios detect hallucinations in one forward pass	Chuang et al. (2024a)	Extend to full layer-wise Shannon entropy sweep on 1B- 3B models
Factual knowledge concentrated in middle-to-upper transformer layers	Chuang et al. (2024b)	Profile entropy per layer, check alignment with DoLa's factual-injection layers

Spectral attention features are current SOTA	Binkowski et al. (2025)	Compare Shannon entropy vs Laplacian eigenvalue features on same test sets
Output entropy over meaning clusters beats token-level perplexity	Farquhar et al. (2024)	Implement semantic entropy as probability baseline, compare against attention entropy
Unsupervised real-time detection achievable from internal states	Su et al. (2024)	Adopt their real-time objective and latency protocol to validate 15% overhead constraint
Detection results overstated when evaluated with ROUGE	Janiak et al. (2025)	Use LLM-as-Judge evaluation, report AUROC, F1, ECE, and latency
Medical and legal hallucination unsolved even for SOTA models	Pandit et al. (2025); Hou et al. (2024)	Test zero-shot cross-domain transfer to both domains to quantify performance degradation

5. Annotated Bibliography:

All sources are peer-reviewed, published 2024–2025, and directly aligned with the project's three experimental tasks and evaluation design.

No	Authors	Title	Venue/Year	Type	Cluster
1	Chuang et al.	Lookback Lens: Detecting and Mitigating Contextual Hallucinations Using Only Attention Maps	EMNLP 2024	Core	Attention
2	Chen et al.	INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection	ICLR 2024	Core	Internal-state
3	Han et al.	Simple Factuality Probes Detect Hallucinations in Long-Form NLG	EMNLP Findings 2025	Core	Probe
4	Farquhar et al.	Detecting Hallucinations in LLMs Using Semantic Entropy	Nature 2024	Core	Probability
5	Su et al.	Unsupervised Real-Time Hallucination Detection Based on Internal States of LLMs	ACL Findings 2024	Core	Consistency
6	Binkowski et al.	Hallucination Detection in LLMs Using Spectral Features of Attention Maps	EMNLP 2025	Core	Attention
7	Chuang et al.	DoLa: Decoding by Contrasting Layers Improves Factuality in LLMs	ICLR 2024	Core	Layer-wise
8	Bang et al.	HalluLens: LLM Hallucination Benchmark	ACL 2025	Core	Benchmark
9	Sahoo et al.	A Comprehensive Survey of Hallucination in Foundation Models	EMNLP Findings 2024	Core	Survey
10	Nikitin & Marttinen	Kernel Language Entropy: Fine-grained Uncertainty Quantification for LLMs	NeurIPS 2024	Core	Probability
11	Janiak et al.	The Illusion of Progress: Re-evaluating Hallucination Detection in LLMs	EMNLP 2025	Supporting	Evaluation
12	Kulkarni et al.	Evaluating Evaluation Metrics: The Mirage of Hallucination Detection	EMNLP Findings 2025	Supporting	Evaluation
13	Pandit et al.	MedHallu: A Comprehensive Benchmark for Detecting Medical Hallucinations in LLMs	EMNLP 2025	Supporting	Medical

14	Hou et al.	Gaps or Hallucinations? Scrutinizing Machine-Generated Legal Analysis	NLLP @ EMNLP 2024	Supporting	Legal
15	Rawte et al.	Tutorial Proposal: Hallucination in Large Language Models	LREC-COLING 2024	Supporting	Background

6. Full Reference List (APA):

- [1] Bang, Y., Ji, S., Schelten, A., Hartshorn, P., Fowler, T., Zhang, L., Cancedda, N., & Fung, P. (2025). HalluLens: LLM hallucination benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics. <https://aclanthology.org/2025.acl-long.1176/>
- [2] Binkowski, M., Janiak, M., Sawczyn, A., Gabrys, B., & Kajdanowicz, T. (2025). Hallucination detection in LLMs using spectral features of attention maps. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. <https://aclanthology.org/2025.emnlp-main.1239/>
- [3] Chen, H., Liu, R., Chen, D., Gu, Q., Wu, Y., Tao, C., Fu, J., & Ye, J. (2024). INSIDE: LLMs' internal states retain the power of hallucination detection. In *Proceedings of the 12th International Conference on Learning Representations*. OpenReview.net. https://proceedings.iclr.cc/paper_files/paper/2024/hash/0d1986a61e30e5fa408c81216a616e20-Abstract-Conference.html
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- [5] Chuang, Y., Xie, Y., Luo, H., Kim, Y., Glass, J., & He, P. (2024b). DoLa: Decoding by contrasting layers improves factuality in large language models. In *Proceedings of the 12th International Conference on Learning Representations*. OpenReview.net. https://proceedings.iclr.cc/paper_files/paper/2024/hash/edc36117f795ca52a0cbf6a7b3882859-Abstract-Conference.html
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- [9] Janiak, M., Binkowski, M., Sawczyn, A., Gabrys, B., Shwartz-Ziv, R., & Kajdanowicz, T. (2025). The illusion of progress: Re-evaluating hallucination detection in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. <https://aclanthology.org/2025.emnlp-main.1761/>
- [10] Kulkarni, A., Zhang, X., Moniz, J. R. A., Ge, T., Tseng, B., Piraviperumal, D., Swayamdipta, S., & Yu, H. (2025). Evaluating evaluation metrics: The mirage of hallucination detection. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics. <https://aclanthology.org/2025.findings-emnlp.1035/>

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- [12] Pandit, A., Xu, M., Hong, P., Wang, J., Chen, M., Xu, Y., & Ding, J. (2025). MedHallu: A comprehensive benchmark for detecting medical hallucinations in large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. <https://aclanthology.org/2025.emnlp-main.143/>
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<https://aclanthology.org/2024.lrec-tutorials.11/>
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7. Conclusion:

A review of 15 peer-reviewed papers from 2024-2025 yielded three main findings. The most promising route towards efficient hallucination detection is via internal-state detection methods, especially those derived from attention distributions, which operate in one forward pass. Second, the 1B-3B model scale is a real research gap: all published attention-based work has only been evaluated on 7B+ parameter models. Third, no paper has conducted a layer-by-layer analysis of attention entropy as a hallucination signal, even though DoLa (Chuang et al., 2024b) offers a theoretical basis to expect such analysis to be informative.

The project design was dictated by the literature itself: layer-by-layer entropy profiling was inspired by the DoLa paper, selection of four complementary evaluation metrics by Janiak et al. (2025) and Kulkarni et al. (2025), medical and legal domains as the most challenging and practically relevant targets for cross-domain evaluation by Pandit et al. (2025) and Hou et al. (2024), and a spectral attention method as the direct state-of-the-art comparison for Shannon entropy features at small model scale by Binkowski et al. (2025).